

SUMMARY

Full-stack software engineer with a background in TypeScript, React, and Node.js. Spent the last few years contributing to Swell, an open-source API testing tool, with a focus on security hardening, test infrastructure, and modernizing legacy architecture. Previously co-founded and led engineering at Mission Coders, building a children's coding education platform from the ground up. Focused on shipping things that are reliable, well-tested, and easy for other engineers to work in. Currently on a health-related career break while continuing technical projects and preparing to return to engineering work.

SKILLS

Languages: TypeScript, JavaScript (ES6), Python
Frontend: React, Redux, Zustand, HTML5, CSS (Sass), Tailwind CSS, Material UI, DaisyUI, Solid.js, Electron.js, webpack, Vite
Backend: Node.js, Express, WebSocket
Databases: PostgreSQL, MongoDB
Infrastructure & Cloud: AWS (EC2, S3, Aurora, Elastic Beanstalk, VPC), Docker, Kubernetes, GitHub Actions
Testing: Jest, Playwright, Mocha, Chai, Vitest
Tools: Git, Cloudinary

EXPERIENCE

Swell · Software Engineer

April 2023 – February 2025

- Implemented Content Security Policy (CSP) in Swell using nonce and SHA-256-based hashing to secure backend protocol testing, mitigate cross-site scripting (XSS) and code injection attacks, and protect user data without impacting application performance
- Reduced technical debt by refactoring legacy Redux architecture: migrated to Redux Toolkit, implemented type-safe hooks, expanded TypeScript coverage, removed deprecated functions and standardized state management to improve maintainability and code quality
- Developed a React-based binary file upload feature using Test-Driven Development (TDD) with Jest, enabling validation of backend server routes across different file types and payload sizes to expand API testing capabilities
- Increased overall test coverage by 15% by repairing failed tests and authoring new unit and integration tests using Jest, Mocha, Chai and Playwright; improved test reliability with Jest mocking and Chai semantic assertions
- Improved application reliability and stability by identifying and resolving security vulnerabilities in API Simulation Tool through root cause analysis and Node.js debugging, preventing crashes and runtime failures.
- Standardized CSS and Material UI theming across the application by unifying CSS classes, implementing Material UI theming architecture, and adding dark mode support to ensure a consistent and accessible user experience.

Mission Coders, LLC · CTO, Co-Founder

May 2018 – December 2022

- Transitioned from an on-premise monolithic server to a scalable AWS infrastructure with Elastic Load Balancing (ELB-ALB) and Elastic Compute Cloud (EC2), reducing resource overhead by 40%, enabling auto-scaling based on traffic patterns, and improving reliability and cost-efficiency across multiple regions
- Designed and implemented a scalable database solution using Amazon Aurora RDS to manage student and course registration; utilized Aurora's auto-scaling capabilities and read replicas to accommodate fluctuating traffic and high read workloads and achieving 99% availability
- Led a six-member team to develop a children's programming learning platform using JavaScript, Node.js, Express, TypeScript, and Redux; integrated Scratch to deliver an engaging, interactive educational tool, completing the project 5 weeks ahead of the deadline

Quark Virtual Services · Technical Manager, Co-Founder

April 2013 – October 2022

- Streamlined deployment for managed and unmanaged Virtual Private Servers (VPS) using KVM and OpenVZ; applied infrastructure-as-code principles and configuration management (Ansible, Puppet) to create reusable, version-controlled templates, reducing manual effort and overall deployment time by 15%
- Drove SEO strategy by conducting keyword research, optimizing on-page elements (title tags, meta descriptions, header tags, content), and established an off-page backlink campaign driving 35% organic traffic growth
- Increased conversion rate by 5% through SEO efforts by monitoring performance via Google Analytics, contributing to revenue growth and business objectives

PROJECTS

QuickChat – Real-time chat platform | github.com/howardsun-dev/quickchat

- Built a real-time messaging platform using React, Node.js/Express, MongoDB, and Socket.IO with authentication. Implemented Zustand for state management, optimistic UI updates, online/offline presence detection and secure media uploads via Cloudinary. Added request protection and rate limiting with Arcjet and built a responsive UI with Tailwind CSS and DaisyUI

Home Automation – Engineering Project

- Engineered a home automation system using Raspberry Pi and Arduino, programmed in C++ for low-level control and performance. Integrated subsystems like automatic blind control with precise timing and PWM. Optimized memory and code execution for smooth operation, increasing home energy efficiency by 10%

EDUCATION

Central Connecticut State University – B.S. Electronic Technology

2012

INTERESTS

Diving into Korean and Russian languages; geeking out over all things Sci-Fi, especially Dune or Star Trek; exploring new cities and embracing nature while camping